

McNamara Analysis

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The composite normative standards used in the McNamara analysis were derived from three sources: lateral cephalograms of the children comprising the Bolton standards; selected values from a group of untreated children from the Burlington Research Center; and a sample of young adults from Ann Arbor, Michigan, with good-to-excellent facial and dental configurations as selected by the author and co-workers.

On examination of the dentofacial skeleton, one must differentiate between the skeletal and dentoalveolar components of a malocclusion. In a normal, well-balanced occlusion, the skeletal and dentoalveolar components of the jaws are well related to each other (Fig 10-1, *a*). A Class II-type malocclusion characterized by a protrusive maxilla (skeletal prognathism) is represented in Fig 10-1, *b*. Because of the protrusive maxilla, the attached dentoalveolar portion has also been carried forward. This type of malocclusion may be treated effectively by extraoral traction in young patients, or by Le Fort I osteotomy, or in certain cases by anterior maxillary osteotomy in adults. Some clini-

cians may elect to camouflage the skeletal discrepancy by extracting teeth and retracting incisors.

Figure 10-1, *c* represents a similar Class II dental condition. The relationship of the maxilla to the mandible is satisfactory; the incisor overjet in this instance is due to the dentoalveolar protrusion. This type of malocclusion can be treated most easily by dental extraction. In certain instances, both skeletal and dental protrusion can contribute to the overall condition. Because the etiology of the condition influences the treatment strategy, it is important to differentiate between skeletal and dentoalveolar abnormalities.

The McNamara analysis divides the craniofacial skeletal complex into five major sections:

1. Maxilla to cranial base
2. Maxilla to mandible
3. Mandible to cranial base
4. Dentition
5. Airway

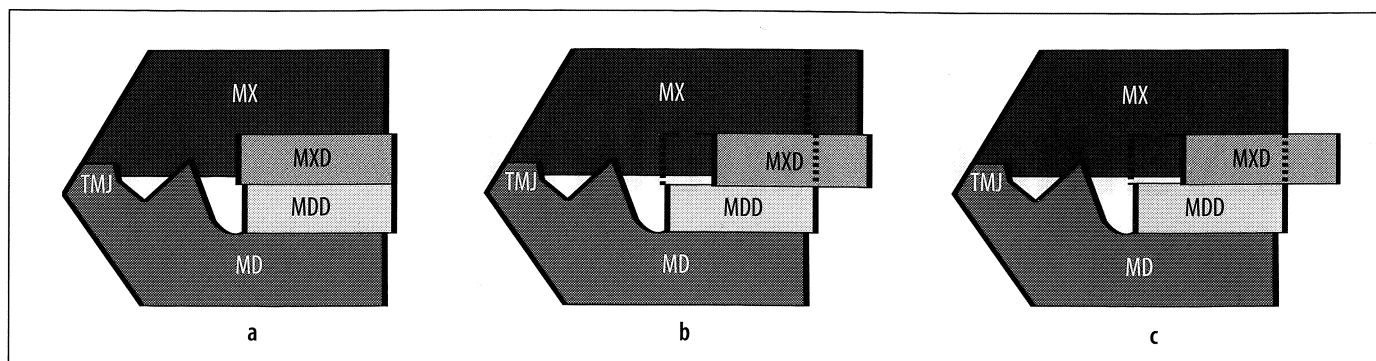


Fig 10-1 (a) Skeletal and dental components of the face in normal occlusion according to McNamara. (b) Maxillary skeletal protrusion. (c) Maxillary dentoalveolar protrusion. The teeth in (b) and (c) are protrusive. In (b) the teeth have been carried forward by the protrusive skeletal maxilla. In (c), the skeletal maxilla is normally positioned; only the dentoalveolus is protrusive. MX = skeletal maxilla; MXD = maxillary dentoalveolus; MDD = mandibular dentoalveolus; MD = mandible; TMJ = temporomandibular joint.



Fig 10-2 Nasolabial angle. The ideal value is 102 ± 8 degrees for adults, female and male.

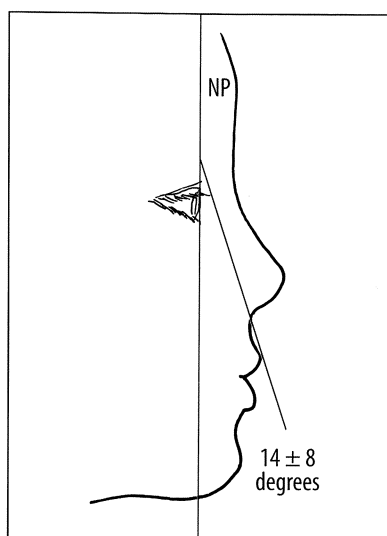


Fig 10-3 The ideal cant of upper lip in adult women is 14 ± 8 degrees; in adult men the ideal value is 8 ± 8 degrees. NP = nasion-perpendicular.

Maxilla to Cranial Base

The position of the maxilla in the skull should be assessed first by clinically observing the soft tissue profile and then by comparing various lateral cephalometric measurements to normative standards.

Soft tissue evaluation

The nasolabial angle and the cant of the upper lip should be examined. The nasolabial angle is formed by drawing a

line tangent to the base of the nose and a line tangent to the upper lip (Fig 10-2). The average nasolabial angle in adult men and women with well-balanced jaws is 102 degrees (SD, 8 degrees). An acute nasolabial angle may be a reflection of the dentoalveolar protrusion, but it also can occur because of the orientation of the base of the nose.

The cant of the upper lip is evaluated by constructing an angle using a line tangent to the upper lip and the nasion-perpendicular (Fig 10-3). The nasion-perpendicular is a vertical line drawn perpendicular to Frankfort horizontal (FH) through nasion. (Note that FH extends from the superior aspect of the external auditory meatus [anatomic portion] to the inferior border of the orbit of the eye [orbitale].

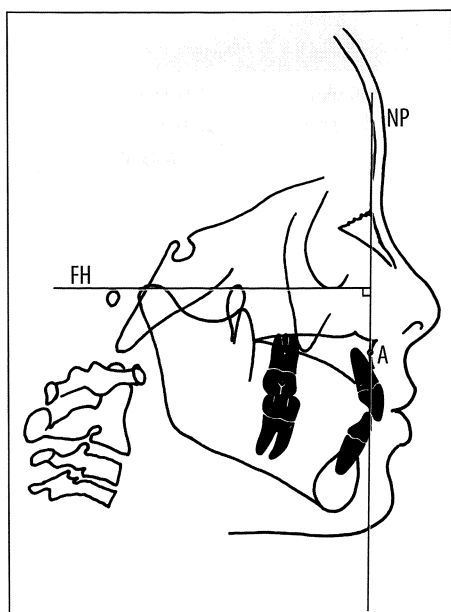


Fig 10-4 In a well-balanced face, nasion-perpendicular (NP) is within 1 mm of point A.

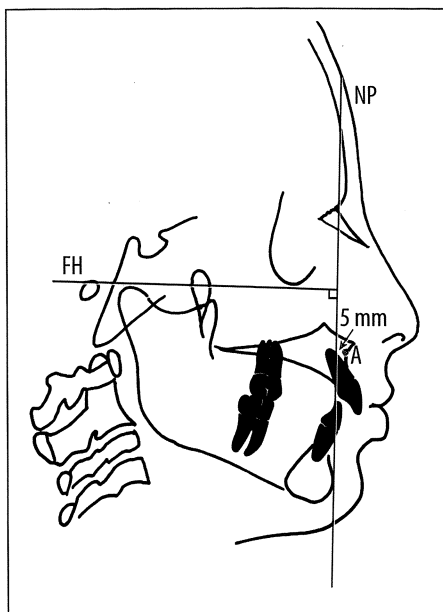


Fig 10-5 Maxillary skeletal protrusion as indicated by the 5-mm distance from point A to nasion-perpendicular (NP).

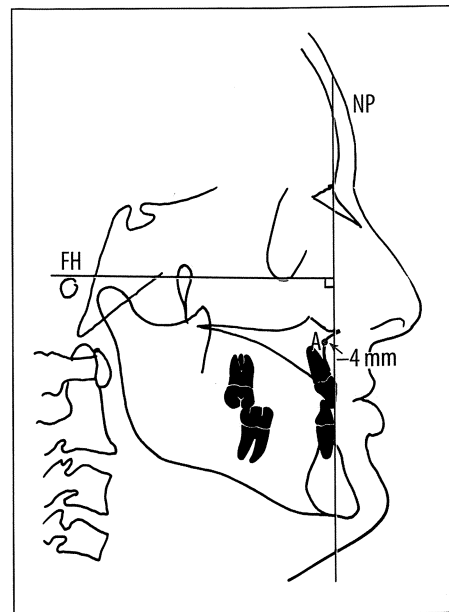


Fig 10-6 Maxillary skeletal retrusion as indicated by the -4-mm distance of point A to nasion-perpendicular (NP).

Machine porion, the top of the ear rods of the cephalostat, is not used because the latter can be 10 mm or more away from anatomic porion.) The angle should be about 14 degrees (SD, 8 degrees) in women and 8 degrees (SD, 8 degrees) in men.

Hard tissue evaluation

To determine the anteroposterior orientation of the maxilla relative to the cranial base, the linear distance is measured between nasion-perpendicular and point A (the posteriormost point of the anterior contour of the maxilla).

An anterior position of point A is a positive value, and a posterior position of point A is a negative value. In well-balanced faces, this measurement is 0 mm in the mixed dentition and 1 mm in adults (Fig 10-4). Figures 10-5 and 10-6 are examples of maxillary skeletal protrusion and retrusion by an amount of 5 mm and -4 mm, respectively.

Maxilla to Mandible

Anteroposterior relationship

A linear relationship exists between the effective length of the midface and that of the mandible (Fig 10-7). Midfacial length is measured from condylion to point A. The effective length of the mandible is measured from condylion to anatomic gonion. Any given effective midfacial length corresponds to an effective mandibular length within a given range (Table 10-1).

It must be stressed that the effective lengths of the midface and mandible as described in the analysis are not age or sex dependent but are related only to the size of the component parts. Thus the terms *small*, *medium*, and *large* are used rather than *mixed dentition*, *adult female*, and *adult male*. In fact, these terms (eg, *large* and *adult male*) are similar in average values, although there is much individual variation in size

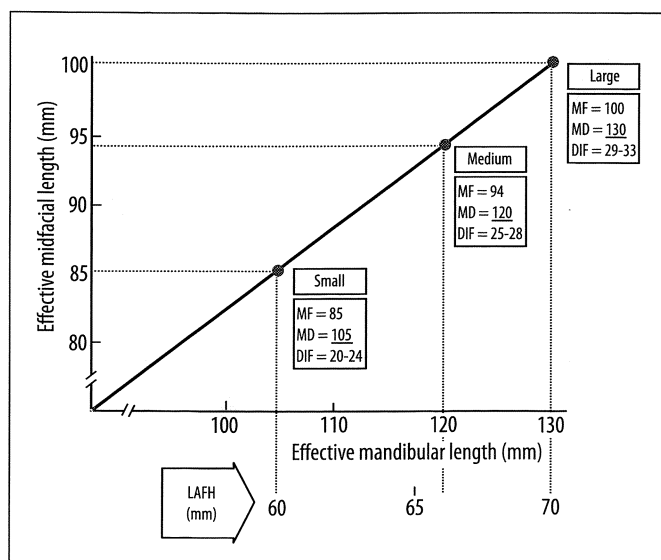


Fig 10-7 The relationship between effective midfacial length and effective mandibular length. This relationship generally is linear and depends on the size rather than the age or sex of the individual. MF = midface, MD = mandible, DIF = maxillomandibular differential, LAFH = lower anterior facial height. (Adapted from McNamara and Brudon¹ with permission.)

Table 10-1 Normative Standards in McNamara Analysis

Midfacial length (mm) (Co-A)	Mandibular length (mm) (Co-Gn)	Lower anterior facial height (mm) (ANS-Me)
80	97–100	57–58
81	99–102	57–58
82	101–104	58–59
83	103–106	58–59
84	104–107	59–60
85	105–108	60–62
86	107–110	60–62
87	109–112	61–63
88	111–114	61–63
89	112–115	62–64
90	113–116	63–64
91	115–118	63–64
92	117–120	64–65
93	119–122	65–66
94	121–124	66–67
95	122–125	67–69
96	124–127	67–69
97	126–129	68–70
98	128–131	68–70
99	129–132	69–71
100	130–133	70–74
101	132–135	71–75
102	134–137	72–76
103	136–139	73–77
104	137–140	74–78
105	138–141	75–79

regardless of age or sex. Thus *small*, *medium*, and *large* are the preferred terms when describing these facial relationships.

To determine the maxillomandibular differential, the midfacial length measurement is subtracted from the effective mandibular length. In small individuals, such as those in the mixed dentition stage, this difference should be between 20 and 24 mm. In medium-sized individuals, the maxillomandibular differential should be between 25 and 28 mm; in large individuals, it should be between 29 and 33 mm.

In the event that the discrepancy is greater or smaller than the normative values, the next step is to identify whether the deficiency or excess is in the maxilla or mandible, or a combination of the two. The relationship of point A to nasion-perpendicular provides some indication of the anteroposterior position of the maxilla. Using this measurement in conjunction with the figures listed in Table 10-1, discrepancies in jaw size can be identified. A normal maxillomandibular relationship and variations are shown in Figs 10-8 and 10-9, respectively.

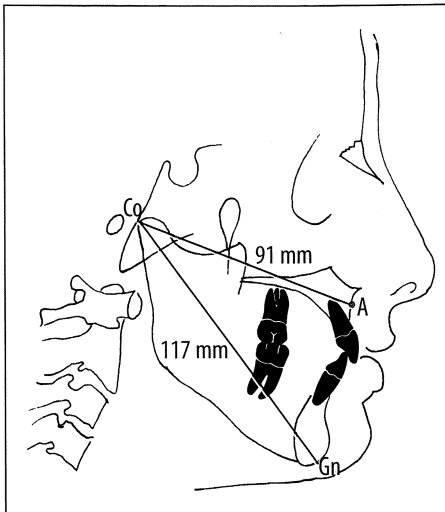


Fig 10-8 Effective midfacial length (Co-A) and mandibular length (Co-Gn) in a well-proportioned face. According to Table 10-1, a midfacial length of 91 mm should have an effective mandibular length of between 115 and 118 mm.

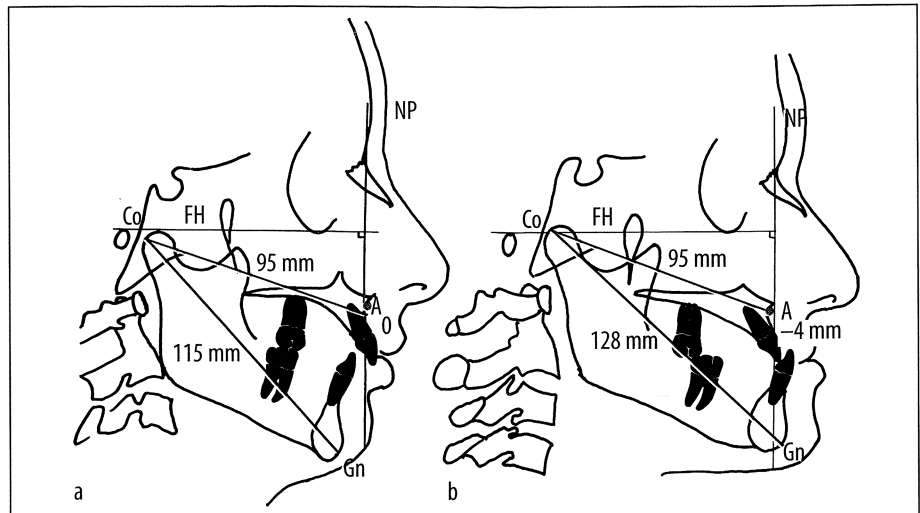


Fig 10-9 Examples of maxillomandibular relationship in medium-sized individuals. (a) Maxillary position is normal (as evidenced by the relative positions of point A and nasion-perpendicular [NP]), mandible is 8-9 mm deficient (see Table 10-1). (b) Skeletal midfacial deficiency of 4 mm (point A is 4 mm posterior to NP); mandibular skeletal excess of 4-5 mm.

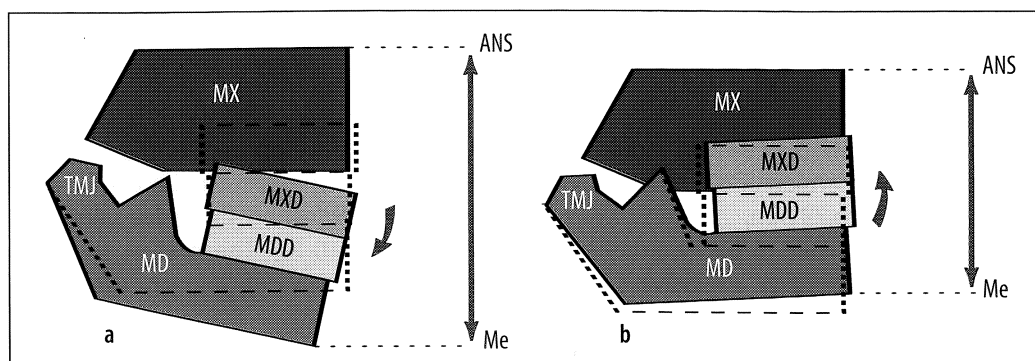


Fig 10-10 (a) Vertical maxillary excess results in downward and backward positioning of the mandible, creating excessive LAFH (ANS-Me). (b) Vertical maxillary deficiency causing an upward and forward positioning of the mandible and deficient LAFH (ANS-Me). MX = skeletal maxilla; MXD = maxillary dentoalveolus; MDD = mandibular dentoalveolus; MD = mandible; TMJ = temporomandibular joint.

Vertical relationship

Vertical maxillary excess can cause a downward and backward rotation of the mandible, resulting in an increase in lower anterior facial height (LAFH) (Fig 10-10, a). Conversely, vertical maxillary deficiency will cause the mandible to rotate upward and forward, thereby reducing the LAFH (ANS-Me, Fig 10-10, b).

LAFH is measured from anterior nasal spine (ANS) to menton (Me). In well-balanced faces, this vertical dimension correlates with the effective length of the midface (condylion–point A; Fig 10-11). The correlations are listed in Table 10-1. An example of excessive LAFH is shown in Fig 10-12.

A midfacial length of 85 mm (categorized as *small*) should be correlated with an LAFH of 60 to 62 mm. LAFH in medium-

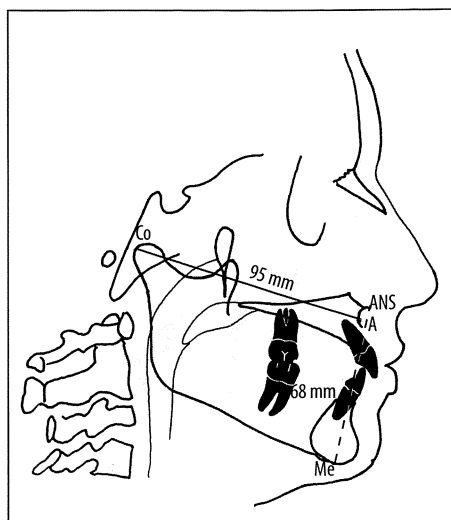


Fig 10-11 In a well-balanced face, effective midfacial length (Co-A) will correlate with LAFH (ANS-Me), according to the normative standard given in Table 10-1.

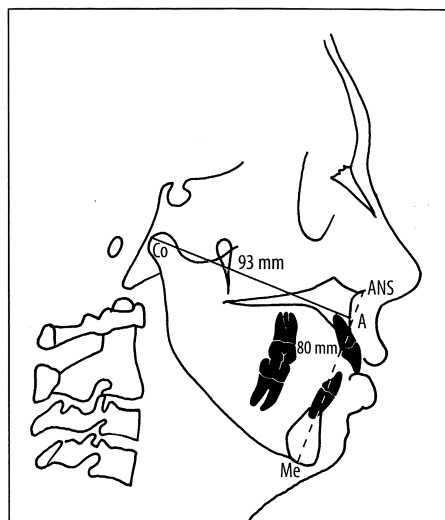


Fig 10-12 Excessive LAFH. With an effective midfacial length (Co-A) of 93 mm, the normal LAFH (ANS-Me) should be 65 to 66 mm (see Table 10-1).

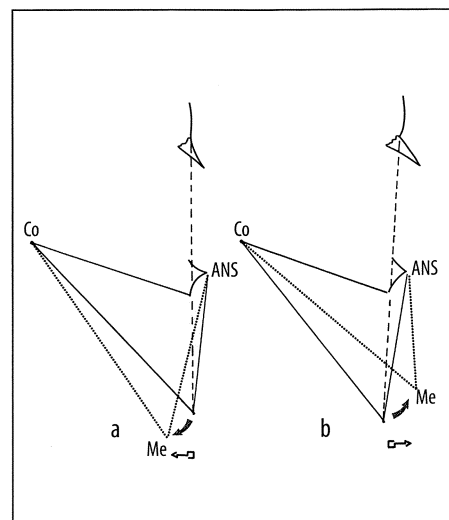


Fig 10-13 The relationship between LAFH (ANS-Me) and forward or backward positioning of the chin. (a) Excessive LAFH will cause the mandible to rotate backward, thereby producing mandibular retrusion. (b) Deficient LAFH will produce forward rotation of the mandible and mandibular protrusion.

sized individuals with a midfacial length of 94 mm should be 65 to 67 mm. A large midfacial dimension of 100 mm should be correlated with an LAFH of 70 to 73 mm.

Figure 10-13 represents the forward or backward effect on the chin attributable to deficient or excessive LAFH, respectively.

The *mandibular plane angle* is the angle between anatomic FH and the line drawn along the lower border of the mandible through constructed gonion (Go) and Me. On average, the mandibular plane angle is 22 ± 4 degrees (Fig 10-14, a). A higher measurement is suggestive of excessive LAFH (Fig 10-14, b), whereas a lesser angle would tend to indicate a deficiency in LAFH. Higher or lower than average mandibular plane angles also can be the result of shorter or longer than average mandibular ramus heights, respectively (or posterior facial height), in which case any conclusions drawn from this single measurement would require other confirmatory measurements.

The *facial axis angle* is formed by a line constructed from the posterosuperior aspect of the pterygomaxillary fissure (PTM) to anatomic gnathion (Gn) and a line perpendicular to the cranial base (represented by a line joining basion [Ba] and nasion [N]). An ideal relationship is when PTM-Gn lies on the perpendicular (0 degrees). If PTM-Gn lies anterior to the perpendicular, the angle is positive, suggesting defi-

cient vertical development of the face. If PTM-Gn lies posterior to the perpendicular, the angle is reported as a negative value, indicating excessive vertical development of the face (Fig 10-15). The greater the absolute value, the greater the vertical deficiency or excess of the face.

Mandible to Cranial Base

The relationship of the mandible to the cranial base is determined by measuring the distance from pogonion (Pog) to N-perpendicular.

In smaller individuals, Pog on average is located 6 to 8 mm posterior to N-perpendicular but moves forward slightly during growth. In an individual with a medium-sized face, such as an adult woman (Fig 10-16, a), Pog is positioned 0 to 4 mm behind the N-perpendicular. In larger individuals, such as adult men, the measurement of the chin position extends from about 2 mm behind to approximately 5 mm forward of the N-perpendicular.

Figure 10-16, b is a tracing of an adult woman exhibiting a severely retrusive mandible and excessive LAFH.

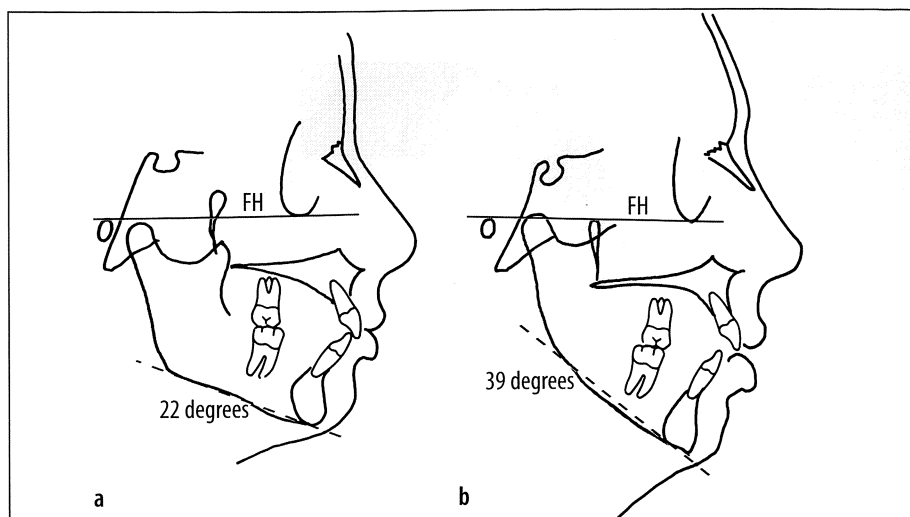


Fig 10-14 (a) Mandibular plane angle (Go-Me to FH) of 22 degrees in an average normal individual. (b) High mandibular plane angle suggestive of excessive LAFH.

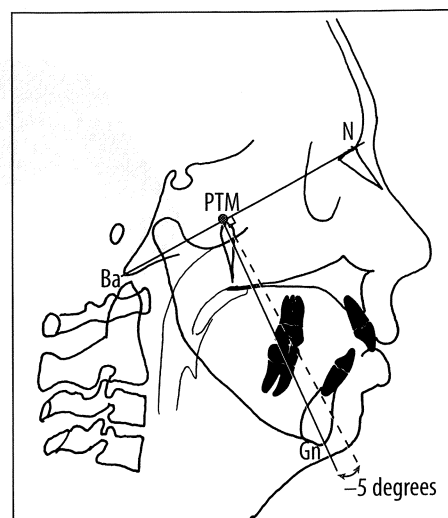


Fig 10-15 Facial axis angle (angle between PTM-Gn and a line perpendicular to Ba-N [dotted line]) of -5 degrees suggests excessive vertical development of the face.

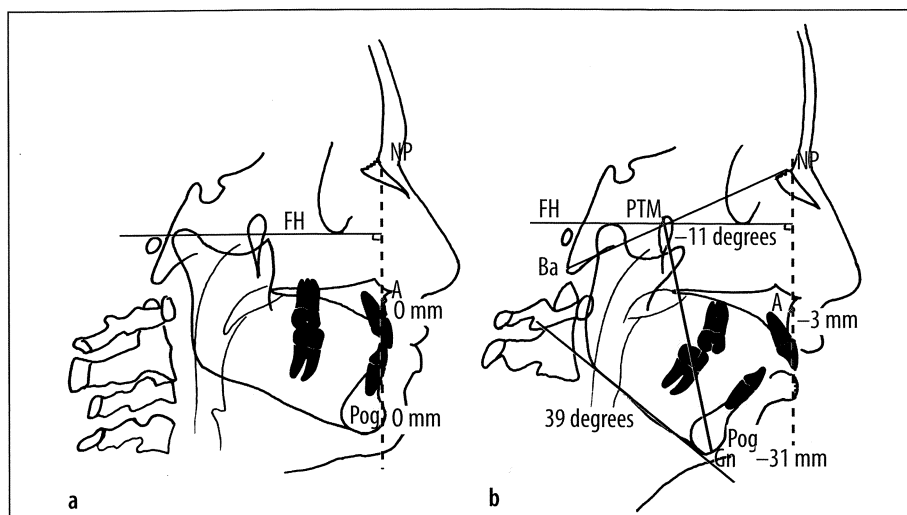


Fig 10-16 Mandible to cranial base measured from Pog to N-perpendicular (NP). (a) Both point A and Pog are located on N-perpendicular, indicating normal maxilla- and mandible-cranial base relationships in an adult woman. (b) Severely retrusive mandible (-31 mm) and mildly retrusive maxilla (-3 mm). High mandibular plane angle (39 degrees) and -11-degree facial axis angle indicate excessive LAFH.

Dentition

In planning orthodontic treatment, be it for orthodontic, orthopedic, or surgical purposes, one must determine the anteroposterior position of both maxillary and mandibular incisors. The reference base to which they are related is described.

Maxillary incisor position

At the outset, it is necessary to know the relationship of the dentition in both jaws to the underlying basal bone. The dentition can be neutral, protrusive as shown in Fig 10-17, *a*, or retrusive as in Fig 10-17, *b*. To determine the position of the maxillary incisors in all such cases, these teeth are measured relative to the position of their respec-

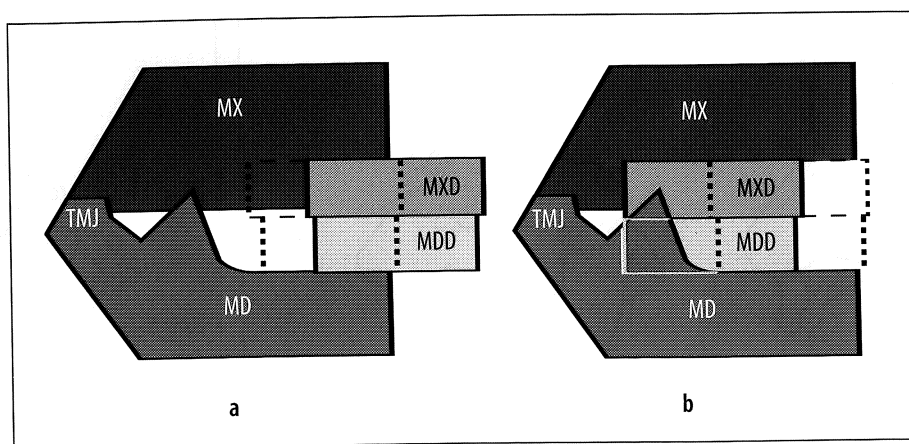


Fig 10-17 Schematic representation of bialveolar protrusion (a) and bialveolar retrusion (b). MX = skeletal maxilla; MXD = maxillary dentoalveolus; MDD = mandibular dentoalveolus; MD = mandible; TMJ = temporomandibular joint.

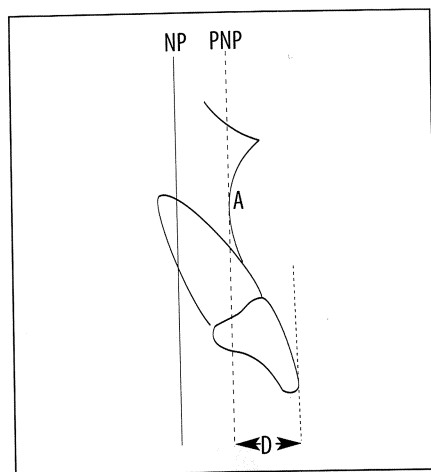


Fig 10-18 Method of determining position of maxillary incisor relative to point A. NP = N-perpendicular; PNP = vertical line constructed parallel to N-perpendicular through point A; D = anteroposterior distance from maxillary incisor to point A (should be 4 to 6 mm).

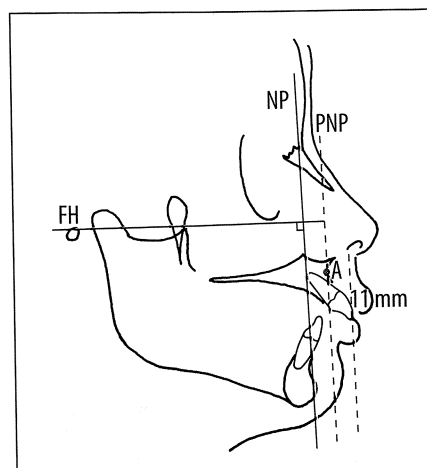


Fig 10-19 Severely protrusive incisors (11 mm) in a protrusive maxilla.

tive bony bases, that is, to the underlying basal bone. To measure the position of maxillary incisors, a vertical line is drawn through point A parallel to N-perpendicular. The distance from point A to the facial surface of the maxillary incisors is measured as shown in Fig 10-18. The ideal distance measured horizontally from point A to the facial surface of maxillary incisors is 4 to 6 mm. The position of the maxillary incisor in Fig 10-16, a is excellent. Figure 10-19 is a tracing of a patient who has severely protruding maxillary incisors (11 mm) in a protrusive maxilla. The incisors in

Fig 10-16, b are moderately repositioned because of a retrusive maxillary base.

Mandibular incisor position

The anteroposterior position of the mandibular incisors must be determined in relation to the mandibular bony base. A differentiation must be made between a Class II malocclusion in which the mandibular dentition is well related in a

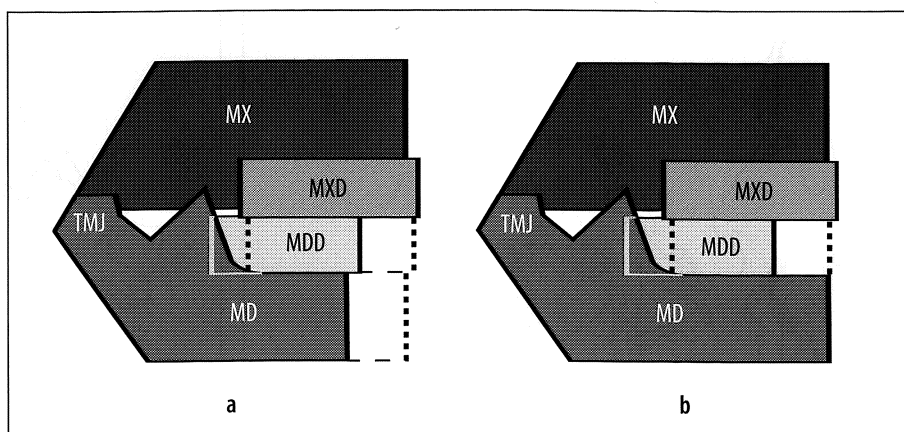


Fig 10-20 (a) Retrusive mandible with incisors positioned normally in relation to the mandibular base. (b) Normal mandible with retruded position of incisors, causing the patient to have a prominent, albeit retrusive, chin. MX = skeletal maxilla; MXD = maxillary dentoalveolus; MDD = mandibular dentoalveolus; MD = mandible; TMJ = temporomandibular joint.

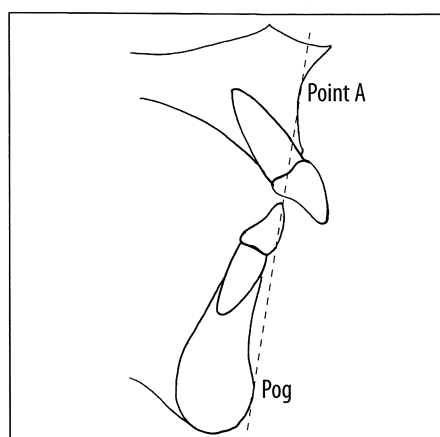


Fig 10-21 A-Pog line. Mandibular incisal edge to A-Pog line distance is measured to determine the anteroposterior position of the mandibular incisor. Maxillary incisors appear labially inclined if the mandible is in a retruded position.

mandible that is retrusive (Fig 10-20, a), and a mandibular dentition that is retrusive in a mandible that is normally positioned in the dentofacial complex (Fig 10-20, b).

To determine the anteroposterior position of the mandibular incisors, the distance is measured between the edge of the incisor and a line drawn from point A to Pog. In a well-balanced face, this distance should be 1 to 3 mm (Fig 10-21).

Numeric estimation of the anteroposterior position of the mandibular incisors is the weakest part of the analysis. For all practical purposes, a subjective evaluation of mandibular incisor position by the clinician is recommend-

ed to determine whether the incisor is positioned well within the mandibular symphysis.

In the assessment of the vertical position of the mandibular incisor, the incisal tip is related to the functional occlusal plane. If the curve of Spee is excessive, a decision must be made as to whether the mandibular incisors should be intruded or the molars erupted. The determining factor is the LAFH. If the LAFH is normal or excessive (determined by relating it to effective midfacial length), the mandibular incisors should be intruded. Should LAFH be deficient, the mandibular incisors should be extruded, or the buccal segments further erupted.

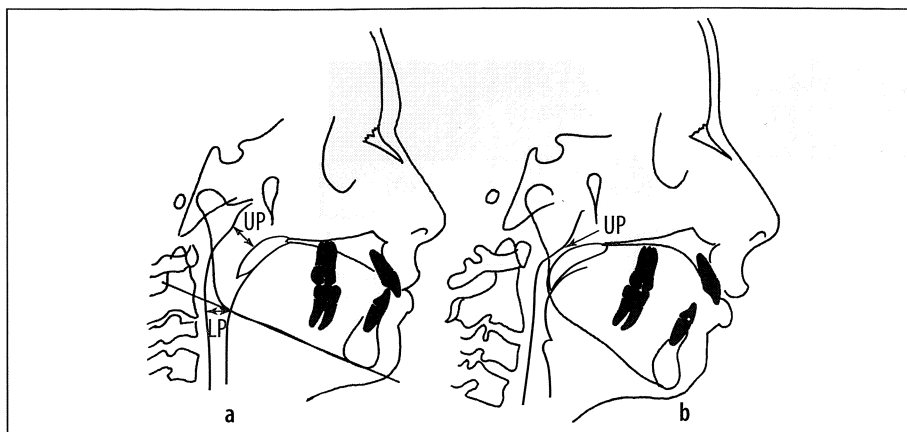


Fig 10-22 (a) Average normal upper pharyngeal airway space (UP), in this instance 15 mm. Lower pharyngeal airway space (LP) measurement is 11 mm. (b) Possible upper airway obstruction; measurement UP is approximately 2 mm.

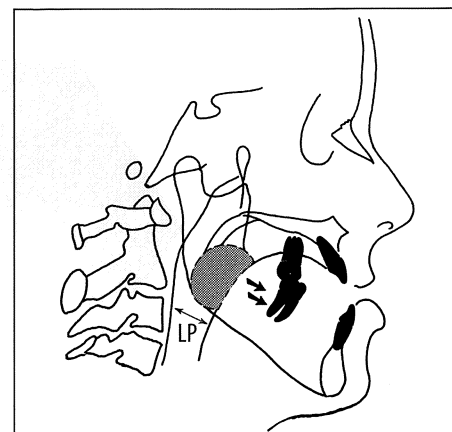


Fig 10-23 Greater than average lower pharyngeal width (LP), suggesting possible anterior positioning of tongue, either as a result of habitual posture or due to tonsillar enlargement.

Airway

For the purpose of this analysis, two measurements are used to examine the possibility of an airway impairment. The relationship between airway obstruction and its effect on craniofacial growth remains unresolved. It must be stressed at the outset that a cephalogram is a two-dimensional representation of a three-dimensional structure. Positive findings observed cephalometrically can serve only as a "red flag"; if indicated, the patient should be examined medically to determine if true airway impairment is present.

Upper pharynx

Upper pharyngeal width is measured from a point on the posterior outline of the soft palate to the closest point on the pharyngeal wall. This measurement is taken on the anterior half of the soft palate outline. The average nasopharynx is approximately 15 to 20 mm in width (Fig 10-22, a). A width of 2 mm or less in this region may indicate airway impairment (Fig 10-22, b). Any suspicion of airway obstruction should be confirmed by an otorhinolaryngologist.

Lower pharynx

Lower pharyngeal width is measured from the point of intersection of the posterior border of the tongue and the inferior border of the mandible to the closest point on the

posterior pharyngeal wall. The average measurement is 11 to 14 mm, independent of age (see Fig 10-22, a).

A smaller than average value for the lower pharynx is of little consequence. An obstruction of the lower pharyngeal area because of a posterior positioning of the tongue against the pharyngeal wall is rare. A greater than average lower pharyngeal width, on the other hand, suggests a possible anterior positioning of the tongue, either as a result of habitual posture or due to tonsillar enlargement (Fig 10-23).

Figure 10-24 is a sample form for clinical application of the McNamara analysis.

Reference

1. McNamara JA Jr, Brudon WL. Orthodontic and Orthopedic Treatment in the Mixed Dentition. Ann Arbor, MI: Needham Press, 1993.

Suggested Reading

- McNamara JA Jr. A method of cephalometric evaluation. Am J Orthod 1984;86:449-469.
- McNamara JA Jr, Brudon WL. Orthodontics and Dentofacial Orthopedics. Ann Arbor, MI: Needham Press, 2001.

McNamara Analysis			
Name of patient _____		Age _____	Sex _____
	Normal	Patient	Comment
1. Maxilla to cranial base Nasolabial angle Cant of upper lip Point A to N-perpendicular	102 ± 8 degrees ♀ 14 ± 8 degrees ♂ 8 ± 8 degrees 0–1 mm	_____ _____ _____	
2. Maxilla to mandible <i>Anteroposterior</i> Maxillary length (Co-A) Mandibular length (Co-Gn) Maxillomandibular differential <i>Vertical</i> LAFH (ANS-Me) Mandibular plane angle (FH-Go-Me) Facial axis angle (PTM-Gn-Ba-N perpendicular)	Varies* Varies* Small 20–23 mm Medium 25–27 mm Large 30–33 mm Varies* 22 ± 4 degrees 0 ± 3.5 degrees	_____ _____ _____ _____ _____ _____	
3. Mandible to cranial base (Pog to N-perpendicular)	Small –8 to –6 mm Medium –4 to 0 mm Large –2 to +5 mm	_____	
4. Dentition Maxillary incisor to point A Mandibular incisor to A-Pog	4–6 mm 1–3 mm	_____ _____	
5. Airway Upper pharynx Lower pharynx	15–20 mm 11–14 mm	_____ _____	
Summary			

Fig 10-24 Sample form for clinical application of McNamara analysis. *Refer to Table 10-1 for normal effective mandibular and maxillary lengths and LAFH.